



Pre-Reads

14-Day AI Accelerator Program

Pre-Read VI

Session Name: Create Voice Agents using Prompting That Talk Like Humans

Table of Contents

Chapter 1: What Are Voice Agents?

- 1.1 From voice assistants to AI-powered voice agents
- 1.2 Why voice agents matter in modern workflows
- 1.3 Voice agents vs. traditional assistants

Chapter 2: Core Building Blocks of Voice Agents

- 2.1 Speech-to-Text (STT)
- 2.2 Natural Language Processing (NLP)
- 2.3 AI Model Processing
- 2.4 Text-to-Speech (TTS)

Chapter 3: Context & Memory in Voice Agents

- 3.1 Why memory makes agents powerful
- 3.2 Session state vs. persistent memory
- 3.3 Example workflow: hospital receptionist bot
- 3.4 Security and compliance considerations

Chapter 4: Voice Agent Tools You Can Explore

- 4.1 VAPI - What Is It and How to Sign Up
- 4.2 Ring AI - What Is It and How to Sign Up
- 4.3 Retell AI - What Is It and How to Sign Up
- 4.4 Real-Time Pipeline

Chapter 5: Designing Voice Workflows & Use Cases

- 5.1 Instruction chaining
- 5.2 Voice + data
- 5.3 Fallbacks when memory is missing
- 5.4 Practical use cases

Chapter 6: Challenges, Best Practices

- 6.1 Accuracy in transcription
 - 6.2 Avoiding context confusion across projects
 - 6.3 Privacy & security for sensitive voice data
 - 6.4 Designing for natural conversation
-

Chapter 1: What Are Voice Agents?

1.1 From voice assistants to AI-powered voice agents

Siri or Alexa is good at basic tasks—setting alarms, playing music, and telling the weather. But that’s where they stop. They don’t manage projects, understand your company context, or chain tasks together.

Voice agents, powered by advanced AI models and delivered through the Voice Agent API (VAPI), go further. They can understand natural speech, remember context across conversations, connect with your tools, and even automate workflows with just your voice. In short, they’re not gadgets - they’re voice-enabled teammates.

1.2 Why voice agents matter in modern workflows

- Hands-free productivity: Perfect for commuting or multitasking.
- Instant access: Get updates and actions without typing.
- Personalisation: They learn your projects and preferences.
- Scalability: Customer-facing agents can carry memory across multiple interactions.

1.3 Voice agents vs. traditional assistants

Traditional Voice Assistant **Voice Agent (MCP + VAPI)**, One-off commands (set timer, play song), Multi-step reasoning and workflows, No memory. Persistent memory with context switching. Limited integrations. Connects with APIs, databases, and custom tools. Robotic responses. Human-like, personalised, and empathetic conversations

Chapter 2: Core Building Blocks of Voice Agents

2.1 Speech-to-Text (STT)

Your spoken words must first be transcribed into text. Modern STT engines are accurate even with accents or background noise, converting the user's spoken input into text format for processing.

2.2 Natural Language Processing (NLP)

Once text is available, the system needs to understand what you mean, identify intent, and extract relevant information. For example:

“Email John the report” → Action = Send email, Recipient = John, File = Report.

2.3 AI Model Processing

This is where reasoning happens. The Large Language Model (LLM) interprets the request, processes the text to understand intent, fetches information, chains steps together, and generates an appropriate response. Model selection directly impacts latency and response quality.

2.4 Text-to-Speech (TTS)

Finally, the AI's text response is converted back to natural speech. Instead of robotic answers, you hear context-aware replies like:

“Got it, I’ve added ‘Budget Review’ under Project Alpha, due Friday.”

Chapter 3: Context & Memory in Voice Agents

3.1 Why memory makes agents powerful

Without memory, every interaction resets. With memory, your voice agent can remember your projects, tone, and previous requests - making it feel like a real teammate. This enables consistent performance and personalised interactions.

3.2 Session state vs. persistent memory

Session state: Temporary memory that exists only within the current conversation (e.g., a single call). It resets once the session ends.

Persistent memory: Long-term memory stored across sessions, lasting days, weeks, or months. It enables ongoing context, personalisation, and learning over time.

3.3 Example workflow: Hospital receptionist bot

You: “I’d like to book an appointment for General Medicine.”

Agent: “Certainly. What is your full name, date of birth, and phone number?”

You: “It’s John Doe, January 1, 1985, and my number is 555-123-4567.”

Agent: “Thank you, Mr Doe. We have slots available on Tuesday at 10 AM or Wednesday at 2 PM. Which do you prefer?”

3.4 Security and compliance considerations

Voice data is sensitive. Agents should respect permissions, keep logs for audits, and ensure compliance with data standards like patient privacy (e.g., HIPAA for healthcare). Robust error handling and fallback mechanisms are crucial.

Chapter 4: Voice Agent Tools You Can Explore

4.1 VAPI - What Is It and How to Sign Up

What Is VAPI?

VAPI (Voice Agent API) is the bridge between voice inputs and AI model outputs. It allows for real-time communication between the spoken command, AI reasoning, and the spoken response. This tool is essential for creating voice-enabled workflows where users can interact with AI agents via voice commands.

Step-by-Step Guide to Sign Up for VAPI:

1. **Visit the VAPI website:** Go to the [VAPI sign-up page](#).
2. **Create an account:** Enter your email, username, and password to create your VAPI account.
3. **Verify your email:** Check your inbox for a verification email and click the link to verify your account.
4. **Complete your profile:** Fill in any additional information required, such as your name and organisation (if applicable).

4.2 Ring AI - What Is It and How to Sign Up

What Is Ring AI?

Ring AI is a conversational AI platform that allows businesses to build and deploy voice agents quickly. It features advanced natural language understanding and is designed for easy integration into existing workflows. It's perfect for creating personalised customer service and support agents.

Step-by-Step Guide to Sign Up for Ring AI:

1. **Visit the Ring AI website:** Go to the [Ring AI sign-up page](#).
2. **Sign up for an account:** Provide your email address and choose a password.

3. **Confirm your email:** You will receive a confirmation email. Click the verification link to activate your account.
4. **Set up your workspace:** After logging in, set up your workspace, which includes specifying the use case for your voice agent.
5. **Explore tutorials:** Ring AI provides tutorials to help you get started with building your voice agents.

4.3 Retell AI - What Is It and How to Sign Up

What Is Retell AI?

Retell AI is a conversational AI platform that specialises in voice interaction and transcription services. It offers powerful voice-to-text conversion capabilities and supports building intelligent, context-aware voice agents for various applications like customer support and personal assistants.

Step-by-Step Guide to Sign Up for Retell AI:

1. **Visit Retell AI's website:** Head over to the [Retell AI sign-up page](#).
2. **Create an account:** Provide your personal information, including email and password, to create a new account.
3. **Confirm your email:** Look for a verification email in your inbox and click the link to confirm your account.
4. **Enter billing information** (if required): Some features may require billing information. Follow the prompts to complete this step.

4.4 Real-Time Pipeline

Here's what happens when you ask a question:

- **You speak your request.**
- **VAPI transcribes speech into text (STT)** (User's spoken input converted to text).
- **The AI processes and reasons with context (LLM)** (Text processed to understand intent and generate a response).

- **VAPI converts the response back into voice (TTS)** (AI's text response converted to natural speech).
- **Network Transport** (Audio data travels over the internet to reach the user).

Latency: Total latency typically ranges from 800 to 1200 milliseconds, depending on model selection and configuration.

Chapter 5: Designing Voice Workflows & Use Cases

5.1 Instruction chaining

Voice agents excel at multi-step instructions and sophisticated workflow automation. For example: “Summarise the report, save highlights, and email the summary to my manager.” This often involves using tools like n8n for complex logic and API integrations.

5.2 Voice + data

You can connect agents to your files and APIs. For instance, upload a PDF (FAQ documents, company policies) and ask: “Summarise the key insights from this document.” This extends the agent's contextual knowledge.

5.3 Fallbacks when memory is missing

If the agent doesn't have past context or a tool is unavailable, the AI can still work with available information, ask clarifying questions, or gracefully degrade to an offline scheduling system. Robust error handling is key here.

5.4 Practical use cases

- **Inbound Appointment Booking Agent:** As demonstrated with "Good Health Hospitals."
- **Resume Screening Agent:** Processing resumes, generating interview questions, and initiating outbound calls.
- **Customer Service Agent:** Handling inquiries with memory and providing consistent information.

- Travel Agent ("Travis"): Created dynamically through AI conversation for travel planning.
 - Voice-triggered automation: (emails, payments, tasks) through custom API integrations.
-

Chapter 6: Challenges, Best Practices

6.1 Accuracy in transcription

Background noise, accents (e.g., Indian accents), and jargon can affect results. Training and fine-tuning multilingual voice models (e.g., ElevenLabs) are essential for improved accuracy.

6.2 Avoiding context confusion across projects

If you juggle multiple projects or manage multiple agents, each agent should maintain its specific personality, goals, and guardrails to prevent tasks from mixing.

6.3 Privacy & security for sensitive voice data

Voice data should be encrypted, and workflows should only access what's permitted. Compliance with regulations (e.g., TRAI in India) and secure API access (OAuth, bearer tokens) are crucial.

6.4 Designing for natural conversation

The goal is human-like flow with empathetic responses. Could you confirm actions, offer next steps, and provide comprehensive guardrails to maintain appropriate interactions, rather than short robotic replies? Iterative refinement of system prompts is key.

Prepare for a hands-on session!

Table of Contents

Chapter 0: Orientation for Builders

0.1 Outcomes: What You'll Be Able to Do

0.2 Prerequisites

0.3 Setup Before the Session

Chapter 1: Understanding the Tech Stack

1.1 What is a Tech Stack?

1.2 The Four Building Blocks of an App

1.3 Key Terms to Know

Chapter 2: Turning Ideas Into Products

2.1 What is an MVP?

2.2 The MoSCoW Method for Prioritising Features

Chapter 3: Introduction to Vibe Coding

3.1 What is Vibe Coding?

3.2 The Role of AI as a Co-Builder

3.3 The EPB Framework – Explore, Plan, Build

Chapter 4: Working with AI Effectively

4.1 AI as a Partner, Not Just a Tool

4.2 Crafting Effective Prompts

4.3 Common Pitfalls to Avoid

Chapter 5: Key Takeaways Before the Session

5.1 How to Prepare for the Session

5.2 Setting Expectations

5.3 Summary Checklist

Chapter 0: Orientation for Builders

0.1 Outcomes: What You'll Be Able to Do

- Understand the core tech terms used in product development.
- Know the building blocks that make up any app or digital product.
- Use AI tools to plan and build a simple MVP, without needing to code.

0.2 Prerequisites

- No prior technical knowledge required — just curiosity and a willingness to experiment.
- A laptop or desktop with internet access.

0.3 Setup Before the Session

- Create accounts on ChatGPT, Claude, and either Bolt or Lovable before the session.
- Come with 1–2 ideas or problems you'd like to build a product around.
- Read through this document — a quick skim is all you need.

Chapter 1: Understanding the Tech Stack

1.1 What is a Tech Stack?

A tech stack refers to the combination of tools and technologies used to build a product. Think of it like choosing ingredients for a recipe — different products need different combinations to work well together.

1.2 The Four Building Blocks of an App

Every web or mobile app is made up of four core components:

- **Front-End:** The part of the app users see and interact with — buttons, screens, and layouts.
- **Back-End:** The logic running behind the scenes — it processes requests and manages how data flows.
- **API:** The connector between the front-end and back-end — it lets different parts of the product communicate.
- **Database:** Where all data is stored — user details, content, history, and anything the app needs to remember.

1.3 Key Terms to Know

- **Tech Stack:** The set of tools used to build a product.
- **Front-End:** What the user sees.
- **Back-End:** What happens behind the scenes.
- **API:** The bridge between front-end and back-end.
- **Database:** Where data lives.

Chapter 2: Turning Ideas Into Products

2.1 What is an MVP?

An MVP (Minimum Viable Product) is the simplest version of your product that solves a core problem. The goal is not to build something perfect — it's to build something that works well enough to test with real users and learn from their feedback.

2.2 The MoSCoW Method for Prioritising Features

Before building, you need to decide what goes into your MVP and what gets left for later. The MoSCoW Method helps with this:

- **Must-have:** Features the product cannot function without.
- **Should-have:** Important additions that add value but aren't critical for launch.
- **Could-have:** Nice to include if time and resources allow.
- **Won't-have:** Intentionally out of scope for now — saved for a future version.

Chapter 3: Introduction to Vibe Coding

3.1 What is Vibe Coding?

Vibe Coding is a way of building real, working products using AI tools — without needing to write code yourself. You focus on the idea, direction, and problem; the AI helps you execute. The result is a working prototype built in a fraction of the usual time.

3.2 The Role of AI as a Co-Builder

Think of AI as a collaborator, not just a tool:

- It can generate code, draft content, suggest structures, and solve problems.
- You provide the direction, context, and goals.
- Together, you turn an abstract idea into something tangible.

3.3 The EPB Framework – Explore, Plan, Build

This is the workflow we'll use during the session to go from idea to prototype:

- **Explore:** Define the problem, your target user, and what the product needs to do at minimum.
- **Plan:** List the features, decide what's in scope for the MVP, and map out the user journey.
- **Build:** Use AI tools to generate, test, and iterate — keeping scope tight and the feedback loop short.

Chapter 4: Working with AI Effectively

4.1 AI as a Partner, Not Just a Tool

Getting good results from AI comes down to how you work with it. Treat it as a collaborator — give it clear direction, review what it produces, and guide it when the output isn't quite right.

4.2 Crafting Effective Prompts

A good prompt has four parts:

- **Role:** Tell the AI what it should act as (e.g. 'Act as a product manager').
- **Task:** Be specific about what you want.
- **Context:** Provide background — what the product is, who it's for, what problem it solves.
- **Format:** Specify how you'd like the output (a list, a table, a short paragraph, code, etc.).

4.3 Common Pitfalls to Avoid

- Vague prompts — the more specific you are, the better the output.
- Using AI outputs without reviewing them first.
- Trying to build too much at once — always stay focused on the MVP.

Chapter 5: Key Takeaways Before the Session

5.1 How to Prepare for the Session

- Think of a product idea or a problem you want to solve.
- Skim through the key terms in Chapter 1 — you don't need to memorise them, just be familiar.
- Set up accounts on ChatGPT, Claude, and Bolt or Lovable.

5.2 Setting Expectations

- This session is about prototyping, not building production-ready apps.
- The focus is on understanding the workflow and learning to collaborate with AI.

- Mistakes are part of the process — experimentation is encouraged.

5.3 Summary Checklist

- Have an idea or problem you want to build around.
- Familiar with the terms: front-end, back-end, API, database, MVP.
- Accounts created on ChatGPT, Claude, and Bolt or Lovable.
- Understanding of the EPB framework: Explore, Plan, Build.
- Open mindset — ready to experiment and iterate.